MAT 115E Introduction to Programming Language

Lab-12/ CRN : 23492

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1 Question 1

Write a C source code that accomplish the followings:

User Defined Function:

- It receives two string arguments along with a parameter specifying the comparison type: case-sensitive or case-insensitive.
- Compares two strings according to the specified comparison type (case-sensitive or case-insensitive) and returns 1 if they are equal; otherwise, returns 0.

Main Function:

- Read the corresponding lines from the files **str1.txt**, **str2.txt**, and **comparison_type.txt**.
- For each line, compare the string from **str1.txt** with the corresponding string from **str2.txt** using the comparison type specified in **comparison_type.txt**. The comparison must be performed using a user-defined function.
- Create a new file named **result.txt**. For each comparison, write `Equal` if the function returns 1; otherwise, write `Not equal` to the corresponding line of the file.

2 Question 2

The following **C FUNCTION** prototype finds all capital letters in a string **str** and saves them in the string **cap**. Write the full body of this function.

void findCapitals(char *str, char *cap)

You should perform following in main function:

- Open the **sentences.txt** file provided on Ninova.
- Read the sentences one by one.
- Identify the capital letters in each sentence.
- Create a file named **capitals.txt** and write the extracted capital letters corresponding to each sentence into this file.

For example:
Input String: `PyThOn Is A PoPuLaR PrOgRaMmInG LaNgUaGe`
Output String: `PTOIAPPLRPORMIGLNUG`

3 Question 3

Consider an integer matrix of size $m \times n$. Write a C program that accomplish the followings:

- Read the **matrix.txt** file. The first line of this file contains two values representing m and n, respectively. The remaining part of the file stores the matrix elements.
- Compute the sum of each row of the matrix.
- Starting from the end of the **matrix.txt** file, write the summation of each row in the following format.

```
Sum of the row 1 is ···.
Sum of the row 2 is ···.
⋮
```